XRPL Validator — Daily Engineering Report · 2026-04-05

Period: 2026-04-05 00:00 to 24:00 US Eastern (2026-04-05 04:00 to 2026-04-06 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 30 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-04-05): 0.94436 (1,189 missed of 21,370). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat: tx verification engine, consensus Phase A, FFI scaffolding** (`741b4e2`, xrpl-validator) Three layers of validator independence work, delivered together: 1. Tx verification engine (deprecated per FFI pivot, retained as scratchpad) - crates/xrpl-node/src/tx_engine.rs: independently verifies transaction effects against rippled's metadata for all 53+ XRPL tx types - Deep verification for Payment (XRP direct), EscrowCreate, PayChan* - Generic fee-verification with pre-state fallback for complex types …
- **feat(ffi): first real libxrpl call — parse mainnet Payment tx** (`8e7a6de`, xrpl-validator) Milestone 2A: STTx deserialization works end-to-end.
- **feat(ffi): first successful xrpl::preflight() call on real mainnet tx** (`9ac92d5`, xrpl-validator) MILESTONE 2B: Full libxrpl preflight pipeline runs end-to-end via FFI.
- **feat(ffi): xrpl::apply() SUCCESS on real mainnet Payment** (`2c9c10e`, xrpl-validator) 🎯 MILESTONE 2C: Full transaction engine runs end-to-end through FFI.
- **feat(ffi): live demo script for visible FFI pipeline** (`2ddeefe`, xrpl-validator) Runs test_shim in a loop, shows live stats: - Stage counters (parse / preflight / apply) with bars - Throughput: iter/sec, calls/sec - Latency: p50/p99/avg per full cycle - Last tx result summary
- **feat(ffi): extract state mutations after apply() — full round trip** (`67eb437`, xrpl-validator) 🎯 MILESTONE 2D: Rust can now get back the exact SLE changes from libxrpl's apply() and commit them to RocksDB. Full round trip complete.
- **feat(ffi): Rust FFI bindings — crate xrpl-ffi with full round trip tests** (`eee986d`, xrpl-validator) 🎯 MILESTONE 2E: Rust can now call libxrpl's tx engine end-to-end.

- **feat(ffi): integrate xrpl-ffi into validator (feature-gated)** (`d8da1f8`, xrpl-validator) Validator can now delegate tx application to libxrpl via the Rust FFI bindings. Feature-gated so default builds are unaffected.
- **feat(ffi): push live mainnet tx through libxrpl in real-time** (`1d7e2eb`, xrpl-validator) The ffi_test server now fetches every new mainnet ledger from rippled and runs each tx through libxrpl's parse + preflight via our FFI.
- **feat(ffi): full apply() on live mainnet tx with RPC-fetched state** (`7d0acd7`, xrpl-validator) Every few seconds, we sample ~3 txs per new mainnet ledger and run them through libxrpl's FULL apply() via FFI. SLE lookups happen synchronously via rippled's ledger_entry RPC (pre-ledger state at seq-1).
- **feat(ffi): 100% mainnet agreement — libxrpl apply in full** (`0f7682f`, xrpl-validator) Three root causes fixed to hit 100% tx-result agreement with mainnet on live feed, up from 55%: 1. Overlay tombstones (LayeredProvider) Deleted SLEs were `remove()`-ed from overlay, so the next lookup fell through to RPC which returned the stale *pre-ledger* version. Now we insert `None` as a tombstone so the fallback is bypassed. Killed OfferCancel/tefBAD_LEDGER (36→0). 55% → 82%.
- **feat(ffi): production observability — Prometheus, Grafana, SLOs** (`00b54ca`, xrpl-validator) Ripple-grade metrics stack around the 100% agreement guarantee: - OpenMetrics /metrics endpoint on port 3778 exposing: * xrpl_ffi_mainnet_agreement_ratio (gauge, SLO target = 1.0) * xrpl_ffi_apply_total{result=tesSUCCESS|tec_claimed|diverged} * xrpl_ffi_apply_duration_seconds histogram (p50/p95/p99) * xrpl_ffi_apply_ter_total{ter}, xrpl_ffi_apply_by_type_total{tx_type} * xrpl_ffi_diverged_total{tx_type,ter} * …
- **fix(ffi): retry + failover against rippled overload** (`9993061`, xrpl-validator) Root cause of 97.26% agreement drop: rippled under memory/load pressure returns plain-text "Server is overloaded" responses (not JSON errors). RpcProvider was parsing these as "no node_binary found" and returning None, which libxrpl interpreted as "this SLE doesn't exist" → terNO_ACCOUNT for accounts that definitely exist on mainnet.
- **refactor(validator): tx_engine → FfiVerifier (libxrpl via FFI)** (`5c0eef0`, xrpl-validator) Replace the 1164-line pure-Rust tx_engine verifier with FfiVerifier, which replays each ledger's tx set through libxrpl (rippled's C++ engine) and tracks mainnet agreement via the same infrastructure ffi_test uses. Deletes tx_engine.rs outright.
- **feat(ffi): Stage 1b — RocksDB-backed SLE provider** (`b1aad92`, xrpl-validator) Add OverlayedDbProvider: 3-tier lookup (in-ledger overlay → rocksdb snapshot of the validator's own state → RpcProvider fallback). Dropping RPC load from ~10 calls/tx to ~0-1 calls/tx once the validator's DB is populated.
- **fix(ffi): OwnedSnapshot eliminates verify race condition** (`0b275be`, xrpl-validator) The bug: Stage 1b's OverlayedDbProvider took a rocksdb snapshot inside spawn_blocking — which could run AFTER ws_sync's process_ledger wrote post-ledger state to the DB. FFI verify then saw POST-state as PRE-state, tried to insert already-existing entries into directories, and libxrpl's ApplyView hit LogicError("dirInsert: double insertion") → std::abort() → process crash.
- **fix(sync): bulk_sync rpc_failover was ignoring XRPL_RPC_URL override** (`8d1d941`, xrpl-validator) rpc_failover() iterated HARDCODED RPC_ENDPOINTS (localhost → xrplcluster → s1.ripple.com). On the validator host (no local rippled), this means every worker falls over to public endpoints that don't serve ledger_data, producing "DONE: 0 objects". The XRPL_RPC_URL env var only affected the initial ledger query, never the worker threads.
- **fix(ffi): disable DB snapshot during catchup — use pure RPC path** (`c2f8f6a`, xrpl-validator)
- **fix(rpc): rpc_request() was ignoring XRPL_RPC_URL env override — used hardcoded localhost** (`1fc4d55`, xrpl-validator)
- **refactor(dashboard): update watch_engine.py for integrated FFI validator** (`e21113b`, xrpl-validator)

- FFI stats now read from validator's /api/engine (ffi_verifier key) instead of separate sidecar at :3778 - Removed deprecated tx_engine section (deleted in code) - Removed "Current Round" and "Top Tx Types" (tied to old engine) - FFI section shows: agreement %, apply count, types, TERs, divergences - Added "State Hash" section with match/mismatch/consecutive counts - FFI_BASE env var no longer needed (stats come …
- **feat(ui): new validator dashboard designed with Google Stitch** (`9613958` , xrpl-validator) Replaces the old validator.html with a production monitoring dashboard generated via Stitch AI. Dark theme, Bloomberg-terminal aesthetic.
- **feat(ui): wire live data into dashboard — 500ms fetch from /api/engine** (`6bcb8ff` , xrpl-validator)
- **feat(ui): live tx-type bars with color-coded fills replacing static chart** (`80eb182` , xrpl-validator)
- **feat(ui): per-ledger tx bars (reset each ledger) + sync progress indicator** (`afdbc36` , xrpl-validator)
- **fix(ui): remove /api/status dependency, infer sync from FFI state** (`ec154d0` , xrpl-validator)
- **feat(ui): show all tx types grayed out, light up when active in current ledger** (`12b4bf8` , xrpl-validator)
- **feat(ui): state lifecycle indicator (sync → build → match) + live object count + round tx stats** (`c190092` , xrpl-validator)
- **feat(ui): descriptive labels + explanations for FFI engine stats** (`6f21057` , xrpl-validator)
- **feat(ui): add WIP notice — dashboard in development, not all links functional** (`77082a0` , xrpl-validator)

Docs and operations

- **docs(ffi): clearer demo — status banner + honest labels** (`a76625e` , xrpl-validator)

Independence program

No independence record for UTC day 2026-04-05; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-05 15:56	—	xrpl-validator	`741b4e2`	feat: tx verification engine, consensus Phase A, FFI scaffolding
04-05 16:32	—	xrpl-validator	`8e7a6de`	feat(ffi): first real libxrpl call — parse mainnet Payment tx
04-05 16:39	—	xrpl-validator	`9ac92d5`	feat(ffi): first successful xrpl::preflight() call on real mainnet tx
04-05 16:46	—	xrpl-validator	`2c9c10e`	feat(ffi): xrpl::apply() SUCCESS on real mainnet Payment
04-05 16:49	—	xrpl-validator	`2ddeefe`	feat(ffi): live demo script for visible FFI pipeline
04-05 16:53	—	xrpl-validator	`a76625e`	docs(ffi): clearer demo — status banner + honest labels
04-05 17:01	—	xrpl-validator	`67eb437`	feat(ffi): extract state mutations after apply() — full round trip
04-05 17:08	—	xrpl-validator	`eee986d`	feat(ffi): Rust FFI bindings — crate xrpl-ffi with full round trip tests
04-05 17:21	—	xrpl-validator	`d8da1f8`	feat(ffi): integrate xrpl-ffi into validator (feature-gated)
04-05 17:25	—	xrpl-validator	`1d7e2eb`	feat(ffi): push live mainnet tx through libxrpl in real-time
04-05 17:30	—	xrpl-validator	`7d0acd7`	feat(ffi): full apply() on live mainnet tx with RPC-fetched state
04-05 19:06	—	xrpl-validator	`0f7682f`	feat(ffi): 100% mainnet agreement — libxrpl apply in full
04-05 19:24	—	xrpl-validator	`00b54ca`	feat(ffi): production observability — Prometheus, Grafana, SLOs
04-05 21:36	—	xrpl-validator	`9993061`	fix(ffi): retry + failover against rippled overload
04-05 21:53	—	xrpl-validator	`5c0eef0`	refactor(validator): tx_engine → FfiVerifier (libxrpl via FFI)
04-05 22:00	—	xrpl-validator	`b1aad92`	feat(ffi): Stage 1b — RocksDB-backed SLE provider
04-05 22:45	—	xrpl-validator	`0b275be`	fix(ffi): OwnedSnapshot eliminates verify race condition
04-05 23:06	—	xrpl-validator	`8d1d941`	fix(sync): bulk_sync rpc_failover was ignoring XRPL_RPC_URL override
04-05 23:11	—	xrpl-validator	`c2f8f6a`	fix(ffi): disable DB snapshot during catchup — use pure RPC path
04-05 23:18	—	xrpl-validator	`1fc4d55`	fix(rpc): rpc_request() was ignoring XRPL_RPC_URL env override — used hardcoded localhost
04-05 23:41	—	xrpl-validator	`e21113b`	refactor(dashboard): update watch_engine.py for integrated FFI validator
04-06 00:28	—	xrpl-validator	`9613958`	feat(ui): new validator dashboard designed with Google Stitch
04-06 00:32	—	xrpl-validator	`6bcb8ff`	feat(ui): wire live data into dashboard — 500ms fetch from /api/engine
04-06 00:44	—	xrpl-validator	`80eb182`	feat(ui): live tx-type bars with color-coded fills replacing static chart

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04-06 00:54	—	xrpl-validator	`afdbc36`	feat(ui): per-ledger tx bars (reset each ledger) + sync progress indicator
04-06 00:57	—	xrpl-validator	`ec154d0`	fix(ui): remove /api/status dependency, infer sync from FFI state
04-06 00:59	—	xrpl-validator	`12b4bf8`	feat(ui): show all tx types grayed out, light up when active in current ledger
04-06 01:05	—	xrpl-validator	`c190092`	feat(ui): state lifecycle indicator (sync → build → match) + live object count + round tx stats
04-06 01:07	—	xrpl-validator	`6f21057`	feat(ui): descriptive labels + explanations for FFI engine stats
04-06 01:12	—	xrpl-validator	`77082a0`	feat(ui): add WIP notice — dashboard in development, not all links functional

Times in UTC. A commit appears on the day it was committed to main or merged into it.